

Content Injection

Adversaries may gain access and continuously communicate with victims by injecting malicious content into systems through online network traffic. Rather than luring victims to malicious payloads hosted on a compromised website (i.e., [Drive-by Target](#) followed by [Drive-by Compromise](#)), adversaries may initially access victims through compromised data-transfer channels where they can manipulate traffic and/or inject their own content. These compromised online network channels may also be used to deliver additional payloads (i.e., [Ingress Tool Transfer](#)) and other data to already compromised systems.^[1]

Adversaries may inject content to victim systems in various ways, including:

- From the middle, where the adversary is in-between legitimate online client-server communications (**Note:** this is similar but distinct from [Adversary-in-the-Middle](#), which describes AiTM activity solely within an enterprise environment)^[2]
- From the side, where malicious content is injected and races to the client as a fake response to requests of a legitimate online server^[3]

Content injection is often the result of compromised upstream communication channels, for example at the level of an internet service provider (ISP) as is the case with "lawful interception."^[3]
^{[1][4]}

ID: T1659

Sub-techniques: No sub-techniques

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Tactics: [Initial Access](#), [Command and Control](#)

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Platforms: Linux, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S1088	Disco	Disco has achieved initial access and execution through content injection into DNS, HTTP, and SMB replies to targeted hosts that redirect them to download malicious files. ^[5]
G1019	MoustachedBouncer	MoustachedBouncer has injected content into DNS, HTTP, and SMB replies to redirect specifically-targeted victims to a fake Windows Update page to download malware. ^[5]

Mitigations

ID	Mitigation	Description
M1041	Encrypt Sensitive Information	Where possible, ensure that online traffic is appropriately encrypted through services such as trusted VPNs.
M1021	Restrict Web-Based Content	Consider blocking download/transfer and execution of potentially uncommon file types known to be used in adversary campaigns.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
		AN0992	Detect suspicious file creations and process executions triggered by browser activity (e.g., injected payloads written to %AppData% or Temp directories, then executed). Correlate network anomalies with subsequent local process creation or script execution.
DET0349	Detection Strategy for Content Injection	AN0993	Detect curl/wget commands saving executable/script payloads to /tmp or /var/tmp followed by execution. Monitor packet captures or IDS/IPS alerts for injected responses or mismatched content types.
		AN0994	Monitor unified logs for processes spawned from Safari or other browsers that immediately load scripts or executables. Detect file drops in ~/Library/Caches or ~/Downloads that execute shortly after being written.

References

1. [Faou, M. \(2023, August 10\). MoustachedBouncer: Espionage against foreign diplomats in Belarus. Retrieved September 1, 2023.](#)
2. [Kaspersky IT Encyclopedia. \(n.d.\). Man-in-the-middle attack. Retrieved September 1, 2023.](#)
3. [Starikova, A. \(2023, February 14\). Man-on-the-side – peculiar attack. Retrieved September 1, 2023.](#)
4. [Budington, B. \(2015, April 2\). China Uses Unencrypted Websites to Hijack Browsers in GitHub Attack. Retrieved September 1, 2023.](#)
5. [Faou, M. \(2023, August 10\). MoustachedBouncer: Espionage against foreign diplomats in Belarus. Retrieved September 25, 2023.](#)